CASE STUDY | INDUSTRIAL ENGINEERING | BEGINNER

Cold-Chain Fulfillment Center — Capacity Scaling (unexpected demand increase)

This is a throughput / capacity problem under a hard budget. Judges are testing whether you find the binding driver before spending, compare three options on the same basis, and show what happens when the numbers move.

THE ANSWER

"Option C — Comprehensive architecture replacement for cold-chain fulfillment center — delivers 13% capacity scaling improvement (target 8.5%) for CAD $676,000 of the CAD $866,000 budget: NPV CAD $2,681,053 at 9.5%, payback 11.1 months."

8.5%	$866K	3 mo	$2.68M
TARGET	**BUDGET CEILING**	**DEADLINE**	**NPV OF ANSWER**
Capacity Scaling improvement	capital, CAD	core implementation	at 9.5% WACC

INSIDE

01 The Challenge
02 The Approach
03 The Options
04 The Results
05 Execution
06 Defence & Submission

Archetype: Operations & engineering improvement · Family: Throughput / capacity · Model: Baseline KPIs → bottleneck/root cause → optimize/simulate → pilot → ROI + KPIs

SITUATION · COMPLICATION · QUESTION

What this case is really asking

This is a throughput / capacity problem under a hard budget. Judges are testing whether you find the binding driver before spending, compare three options on the same basis, and show what happens when the numbers move.

S SITUATION

Cold-Chain Fulfillment Center — Capacity Scaling (unexpected demand increase). Metropolitan public agency; disruption: unexpected demand increase. Capacity Scaling today: 1,190.00 units/day.

C COMPLICATION

Capacity Scaling must move from 1,190.00 units/day to 1,291.15 units/day (8.5%) within 3 months and CAD $866,000, while: installation restricted to weekends; Minimum availability 98.0%.

Q QUESTION

For the cold-chain fulfillment center operated by a metropolitan public agency, plan a capacity upgrade while protecting service quality following unexpected demand increase. Baseline capacity: 1,190 units/day. Deliver at least 8.5% improvement within 3 months under CAD 866,000 capital limit; installation restricted to weekends. Select a technically justified alternative and test what happens if performance falls 15%.

A ANSWER

Option C — Comprehensive architecture replacement for cold-chain fulfillment center — delivers 13% capacity scaling improvement (target 8.5%) for CAD $676,000 of the CAD $866,000 budget: NPV CAD $2,681,053 at 9.5%, payback 11.1 months.

CONSTRAINTS

- installation restricted to weekends
- Minimum availability 98.0%

WHAT THE JUDGES SCORE



SOLUTION MODEL · OPERATIONS & ENGINEERING IMPROVEMENT

How we cracked it

Baseline KPIs → bottleneck/root cause → optimize/simulate → pilot → ROI + KPIs

How do we achieve 8.5% capacity scaling improvement within 3 months and CAD $866,000?

Demand & mix

- Is the demand real and sustained?
- Does product mix drive changeovers?

Capacity at the constraint

- Which stage runs at the highest utilisation?
- Where are OEE losses: availability, speed or quality?

Flow between stages

- Where does WIP pile up or starve?
- Are batch sizes and buffers sized to variability?

HYPOTHESIS (FORMED IN THE FIRST 20 MINUTES)

"The gap is one binding constraint, not total capacity: relieving it delivers the target without rebuilding the whole system."

TOOLKIT

- Theory of Constraints (5 focusing steps)
- OEE = Availability × Performance × Quality
- Little's Law: WIP = Throughput × Flow time
- Discrete-event simulation

QUICK MATH

- Gap: 1,190.00 → 1,291.15 units/day
- Budget per % point of target: CAD $101,882
- Winner cost per % point delivered: CAD $52,000

TIME PLAN FOR A 5-HOUR CRACKING WINDOW

0:00–0:40	0:40–1:50	1:50–3:20	3:20–4:20	4:20–5:00
Read & align	Data & quick math	Options & decision	Deck	Rehearse & Q&A
SCQA, issue tree, first hypothesis	Baseline, gap, vital-few drivers (80/20)	3 costed options, matrix, NPV, sensitivity	Answer-first storyline, action titles	Timed run-through, 20+ questions

SOLUTION LADDER

Four levels of solution, one recommendation

Option C is recommended. It is both the cheapest compliant option and the best value: a dominant choice.

LEVEL 1 · QUICK WIN

Option A

4.6% · $232K

Lowest capital: what a minimal move buys

LEVEL 2 · MINIMUM COMPLIANT

Option C

13% · $676K

Cheapest option that meets the target within budget

RECOMMENDED

LEVEL 3 · VALUE MAXIMISER

Option C

13% · $676K

Highest NPV within budget

RECOMMENDED

LEVEL 4 · STRETCH

Option C

13% · $676K

Largest improvement regardless of budget

RECOMMENDED

DECISION MATRIX (SCORE 0-10, WEIGHTED)

Criterion	Weight	Option A	Option B	Option C (chosen)
Target attainment	30%	3.6	5.4	10.0
Financial return (NPV)	25%	2.0	3.4	10.0
Payback speed	15%	8.2	7.5	8.2
Downside resilience (-15% perf.)	15%	4.6	6.9	10.0
Budget headroom	15%	7.3	4.2	2.2
Weighted total	100%	4.6	5.3	8.6

THE THREE ALTERNATIVES

Option A: Instrumentation and operating procedure changes for cold-chain fulfillment center	$232K · 4.6% · NPV $957K In budget · misses target
Option B: Selective retrofit of critical cold-chain fulfillment center subsystems	$502K · 6.9% · NPV $1.25M In budget · misses target
Option C: Comprehensive architecture replacement for cold-chain fulfillment center	$676K · 13% · NPV $2.68M In budget · meets target

ECONOMICS · PROFIT-DRIVEN

What the recommendation is worth

Run-rate benefit $948K/yr net of $34K/yr O&M; levelized 5.43 CAD per added unit.

$2.68M	119.2%	11.1 mo	4.33x
NET PRESENT VALUE	**IRR**	**PAYBACK**	**BENEFIT / COST**
5 yrs @ 9.5%	annual	monthly cash model	present value

CASH FLOW BY YEAR (BARS) AND CUMULATIVE (LINE)



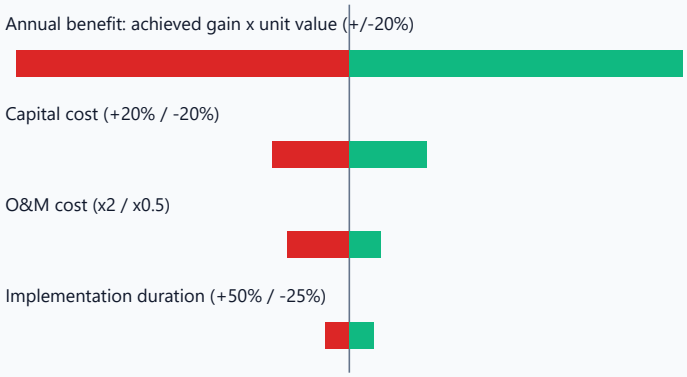
SCENARIOS

Bear (-15% perf., +20% capex)
11.05% · NPV $2.00M · payback 14.9 mo · meets target

Base (approved plan)
13% · NPV $2.68M · payback 11.1 mo · meets target

Bull (+15% perf., -10% capex)
14.95% · NPV $3.28M · payback 9.1 mo · meets target

NPV SENSITIVITY (TORNADO)



30 / 60 / 90 DAYS

From pilot to control

One owner per workstream; every phase ends in a go/no-go gate.

DAYS 1–30

Validate & pilot

- Confirm baseline data and the binding constraint
- Launch pilot of Option C at the constraint
- Name owners; start KPI dashboard

DAYS 31–60

Prove & decide

- Pilot gate: Go/no-go: pilot result within 20% of model, no safety incidents
- Lock fixed-price packages for scale-up
- Train operators; write standard work

DAYS 61–90

Scale & control

- Begin scale-up: comprehensive architecture replacement for cold-chain fulfillment center
- Control chart on the primary KPI
- Review vs target (8.5%) with sponsor

IMPLEMENTATION HORIZONS

Horizon 1 · Month 1 **Baseline validation & pilot**	KPI: Pilot reaches >= 2.5% improvement; data quality verified Gate: Go/no-go: pilot result within 20% of model, no safety incidents	**$135K**
Horizon 2 · Month 2 **Scale-up: Comprehensive architecture replacement for cold-chain fulfillment center**	KPI: >= 6.8% improvement sustained for 4 weeks Gate: Commissioning sign-off and availability at or above constraint	**$406K**
Horizon 3 · Month 3 **Stabilise, optimise & hand over**	KPI: Full target 8.5% reached; O&M within CAD $33,800/yr Gate: Benefit-tracking dashboard owned by operations	**$135K**

TOP RISKS

Delivered improvement below model (Medium / High)

Mitigation: Stage-gated rollout; bear case still yields 11.0% vs 8.5% target
Trigger: Pilot result < 80% of modelled gain

Capital overrun (Medium / Medium)

Mitigation: CAD $190,000 headroom to budget ceiling; fixed-price packages for long-lead items
Trigger: Committed cost > 90% of H2 allocation

Schedule slip / installation window (Medium / Medium)

Mitigation: Pre-fabrication and off-shift installation; pilot proves method before scale-up
Trigger: Critical-path float < 2 weeks

KPIS: BASELINE → TARGET

Daily output

1,190.00 units/day → 1,291.15 units/day

Constraint utilisation

measure in week 1 → set from pilot

OEE at the constraint

measure in week 1 → set from pilot

WIP

measure in week 1 → set from pilot

BEFORE YOU SUBMIT

Judge questions and the final checklist

9 prepared answers, all computed from the model. Answer first, then evidence.

STRESS TEST

What changes when unexpected demand increase persists and the selected design underperforms by 15%?

Bear case (performance -15%, capex +20%): 11.0% improvement, NPV CAD $1,996,871, payback 14.9 months. Target is still met.

ALTERNATIVES

Why not Option A, the cheapest?

It costs CAD $232,000 but delivers 4.6% (misses the 8.5% target); it scores 4.6/10 vs 8.55/10 for Option C.

ASSUMPTIONS

How confident are you in the option estimates?

Source: case datum. We treat them as hypotheses: the Horizon 1 pilot must reach 80% of the modelled gain before scale-up spend.

FINANCIALS

What does it cost to run every year?

O&M of CAD $33,800/yr is already netted out of the CAD $947,537/yr benefit.

FINANCIALS

What is the benefit-to-cost ratio?

4.33 on a present-value basis at 9.5% WACC; levelized 5.43 CAD per added unit.

IMPLEMENTATION

What if implementation takes 50% longer?

NPV falls to CAD $2,630,575; the H1 pilot and fixed-price packages protect the schedule.

PRE-SUBMISSION CHECKLIST

✓ Answer-first executive summary (one sentence, three numbers)	Option C — Comprehensive architecture replacement for cold-chain fulfillment center — delive…
✓ Three options costed on the same basis	decision matrix with weights shown
✓ Recommendation fits the budget	CAD $676,000 of CAD $866,000
✓ Recommendation meets the target	13% vs 8.5%
✓ Financials: NPV, payback, sensitivity	NPV CAD $2,681,053, payback 11.1 months
✓ Bear case tested (-15% performance, +20% cost)	bear NPV CAD $1,996,871
✓ 30/60/90 roadmap with owners and gates	pilot → prove → scale
✓ Top risks with triggers	5 risks logged
✓ Q&A bank of 10+ prepared answers	see judge_QA_preparation.md
! Assumptions validated with real data	replace: Monetisation, Discount rate (WACC), Appraisal horizon, Annual O&M, Benefit during i…
✓ Safety & licensed engineering review noted	MOC + HAZOP before commissioning

WHERE THIS CASE STYLE APPEARS (DIRECTORY-LISTED; WINNERS NOT VERIFIED)

- Boeing Northwest Case Competition · USA
- Colorado State University Supply Chain Case Competition · USA
- Carlson National Undergraduate Supply Chain Case Competition · USA
- University of St. Thomas Supply Chain Case Competition · USA
- Kohler Undergraduate Wisconsin Supply Chain Cup · USA